

# Project - Credit Card Complaints Dashboard

## Introduction:

In today's fast-paced financial ecosystem, customer feedback is one of the most valuable indicators of service quality and operational effectiveness. The Credit Card Complaints Dashboard was developed to systematically track, analyze, and visualize consumer complaints submitted regarding credit card-related issues. Using Tableau, this interactive dashboard provides a holistic view of complaints across time, geography, issue types, company responses, and communication channels. It includes key performance indicators such as the total number of complaints, timely response rates, in-progress case volumes, and trends across weeks and states. A combination of visual tools like sparklines, progress bars, calendar heatmaps, and geospatial maps allows for both high-level summaries and granular insights.

## Steps:

### 1. Load Data:

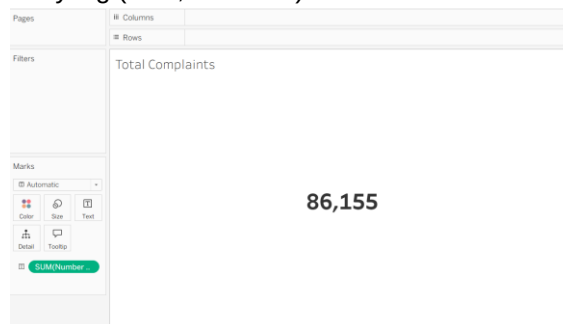
Loaded data by creating connection with 'Credit Card Dashboard.xlsx'

### 2. Sheet 1: Total Complaints

**Objective:** Display total complaint volume.

2.1. Dragged Number of Records to Text → SUM(Number of Records)

2.2. Formatted the text for styling (bold, centered).



### 3. Sheet 2: Rolling 12 Months Complaints

**Objective:** Show complaints received in the past 12 months.

3.1. Dragged 'Date Received' to row → Month

3.2. Created a calculated field to determine the latest date received.

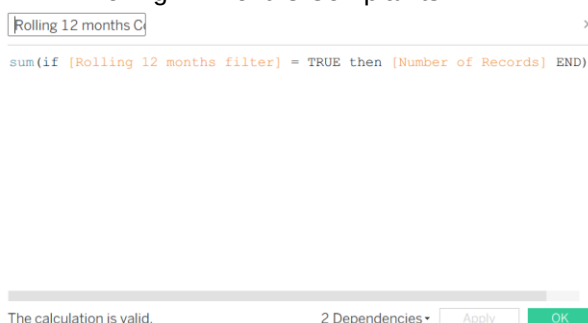


3.3. Created a parameter:

### 3.4. Created another parameter 'Rolling 12 months filter':



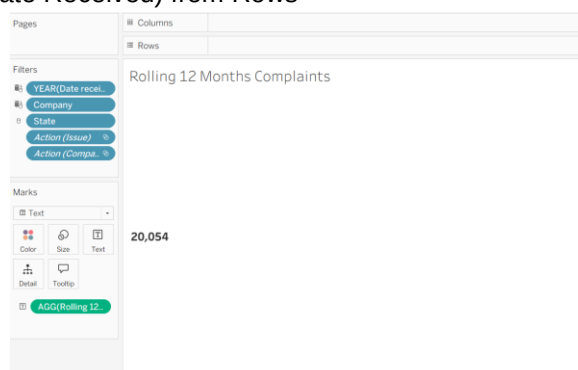
### 3.5. Created another parameter 'Rolling 12 months Complaints':



### 3.6. Dragged 'Rolling 12 months Complaints' to Text

### 3.7. Formatted the label.

### 3.8. Removed MONTH(Date Received) from Rows



## 4. Sheet 3: Complaints Sparkline

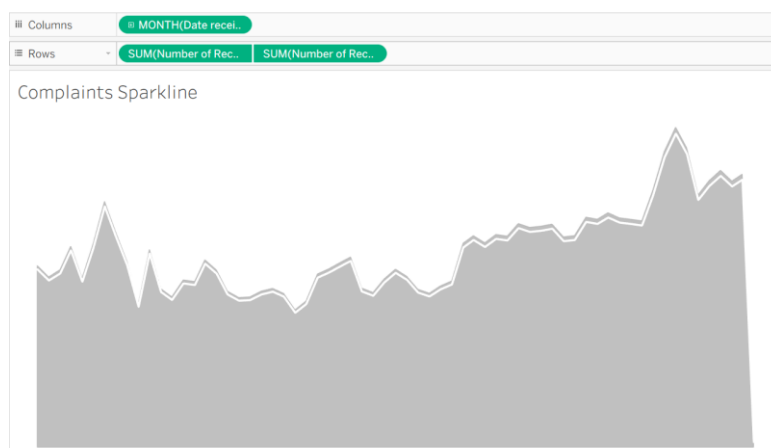
**Objective:** Visualize trend over time.

4.1. Used the 'Date received' field truncated to months on the columns shelf.

4.2. Plotted the sum of 'Number of Records' as an area chart.

4.3. Duplicated the SUM(Number of Records) and overlaid it as a line chart using dual axis.

4.4. Synchronized axes and formatted colors and borders to create a clean sparkline visualization.



## 5. Sheet 4: Timely Response

**Objective:** Total complaints responded to on time.

5.1. Create calculated field:

Timely Response

```
if [Timely response?] = 'Yes' THEN 1 ELSE 0 END
```

The calculation is valid.

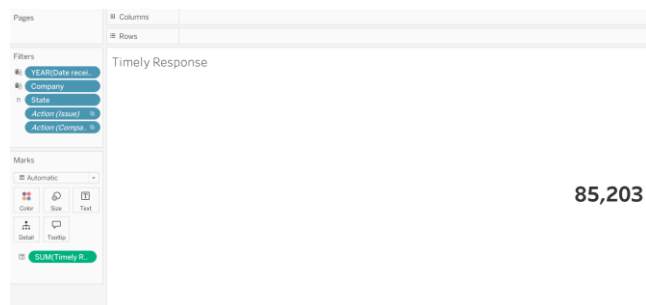
5 Dependencies

Apply

OK

5.2. Dragged it to Text

5.3. Formatted the label



## 6. Sheet 5: Closed %

**Objective:** Show proportion of timely responses out of total.

6.1. Created a calculated field:

Closed %

```
sum([Timely Response])/sum([Number of Records])
```

The calculation is valid.

3 Dependencies

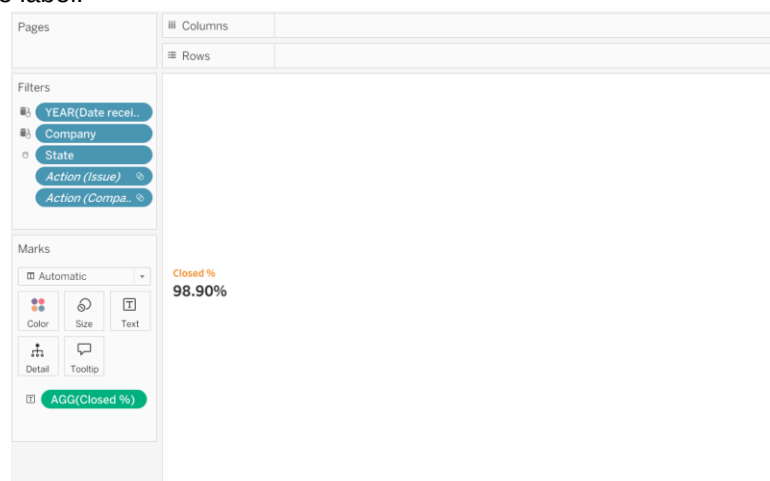
Apply

OK

6.2. Dragged it to Text

6.3. Formatted it as percentage

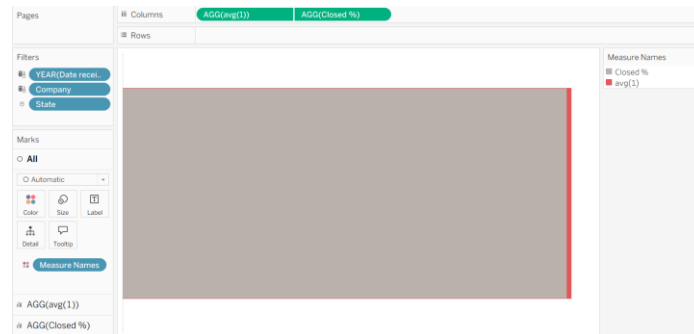
6.4. Formatted the label.



## 7. Sheet 6: Progress Bar

**Objective:** Visual representation of Closed %.

- 7.1. Dragged 'Closed %' to Columns
- 7.2. Double clicked in Columns → avg(1) → Dual Axis
- 7.3. Set both of these Marks as Bar
- 7.4. Edited the colors of Bar



## 8. Sheet 7: In Progress Complaints

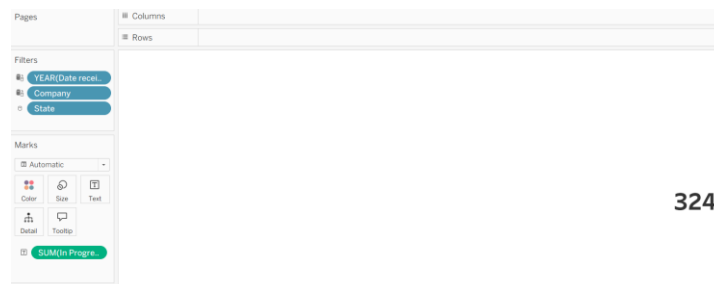
**Objective:** Count of complaints marked "In Progress".

- 8.1. Create calculated field 'In Progress Complaints':



- 8.2. Dragged it into Text

- 8.3. Formatted the label



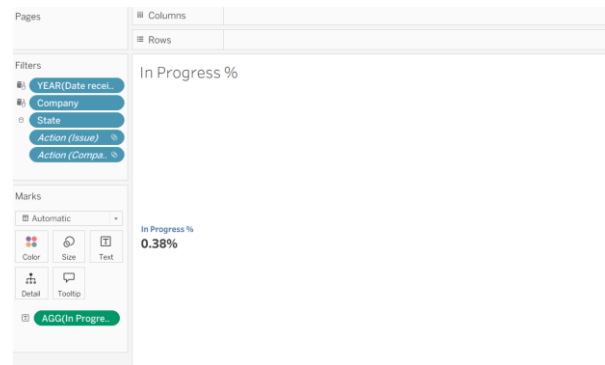
## 9. Sheet 8: In Progress %

**Objective:** Show what percentage of complaints are still open.

- 9.1. Created a calculated field 'In Progress %':



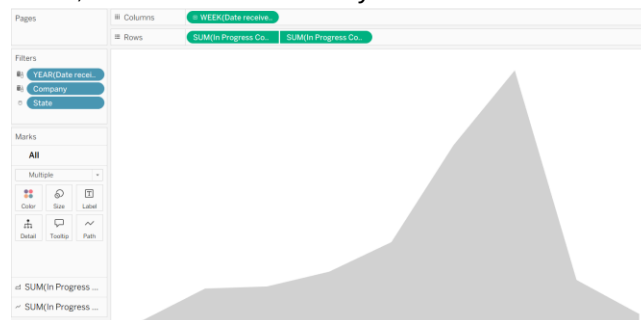
- 9.2. Dragged it to Text
- 9.3. Formatted the type to Percentage
- 9.4. Formatted the label



## 10. Sheet 9: In Progress by Week (Sparkline)

**Objective:** Weekly trend of in-progress complaints.

- 10.1. Dragged 'Date received' to Columns and converted to Week Number.
- 10.2. Dragged 'In Progress Complaints' to Rows and duplicated → Dual Axis
- 10.3. Set one Marks as Line and another as Area.
- 10.4. Synchronized the axes, and formatted for clarity



## 11. Sheet 10: Trend

**Objective:** Show weekly complaint trend.

- 11.1. Created a parameter 'Trend':

The screenshot shows the 'Edit Parameter [Trend]' dialog box. The Name field is 'Trend'. The Data type is 'String' and the Display format is 'week'. The Current value is 'Weekly' and the Value when workbook opens is 'Current value'. The Allowable values are set to 'List'. The Value field has a table with the following data:

| Value   | Display As |
|---------|------------|
| month   | Monthly    |
| week    | Weekly     |
| quarter | Quarterly  |
| year    | Yearly     |
| day     | Daily      |

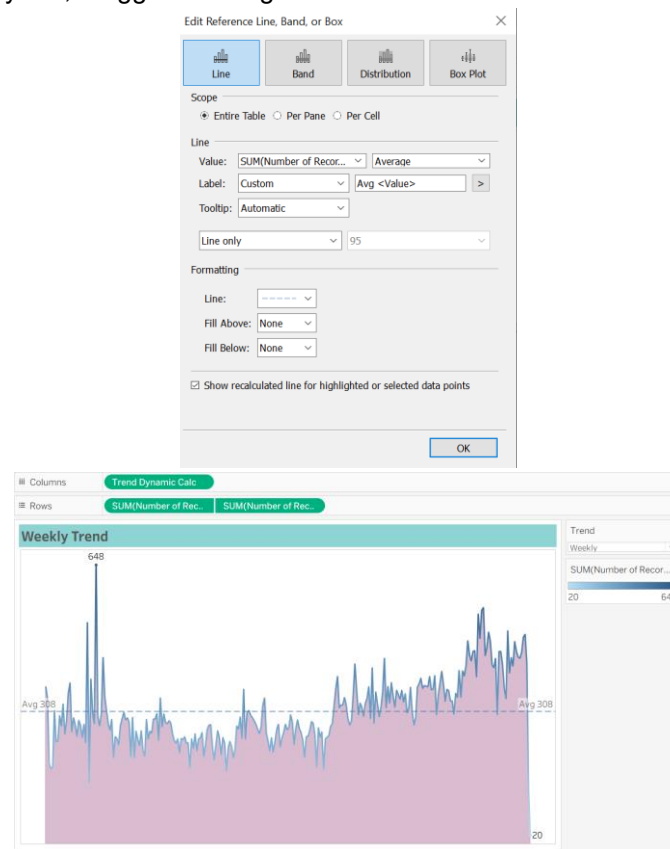
The 'Fixed' radio button is selected. The 'Add values from' dropdown is set to 'Current value'. The 'Remove Selected' button is disabled. The 'Cancel' and 'OK' buttons are at the bottom.

- 11.2. Created a calculated field 'Trend Dynamic Calc':

The screenshot shows the 'Trend Dynamic Calc' calculated field definition. The formula is: `DATE(DATETRUNC([Trend],[Date received]))`. The calculation is valid, and there are 2 dependencies. The 'Apply' and 'OK' buttons are at the bottom.

- 11.3. Dragged 'Trend Dynamic Calc' to Columns and 'Number of Records' to Rows.

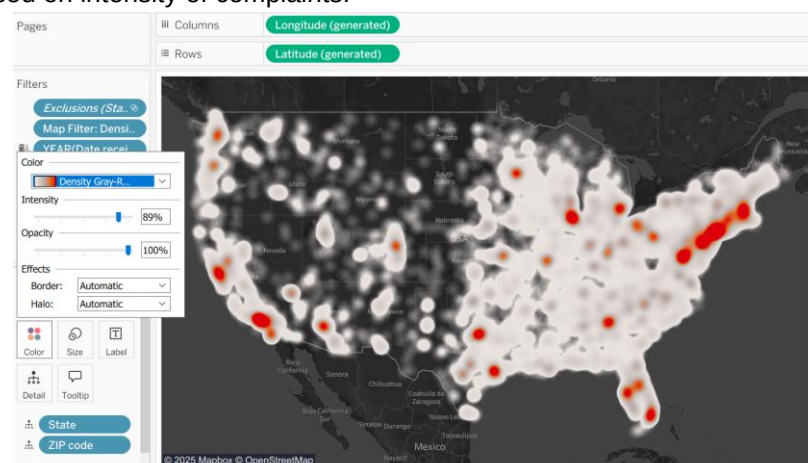
- 11.4. Duplicated SUM(Number of Records) → Dual Axis
- 11.5. Set one Marks as Line and another as Area.
- 11.6. Synchronized the axes, and formatted for clarity
- 11.7. Dragged 'Number of Records' in the Color and Label of the Line Chart. In the label selected 'Min/Max' in 'Marks the Label'
- 11.8. From 'Analytics', dragged 'Average Line' and edited:



## 12. Sheet 11: Density Map

**Objective:** Show concentration of complaints geographically.

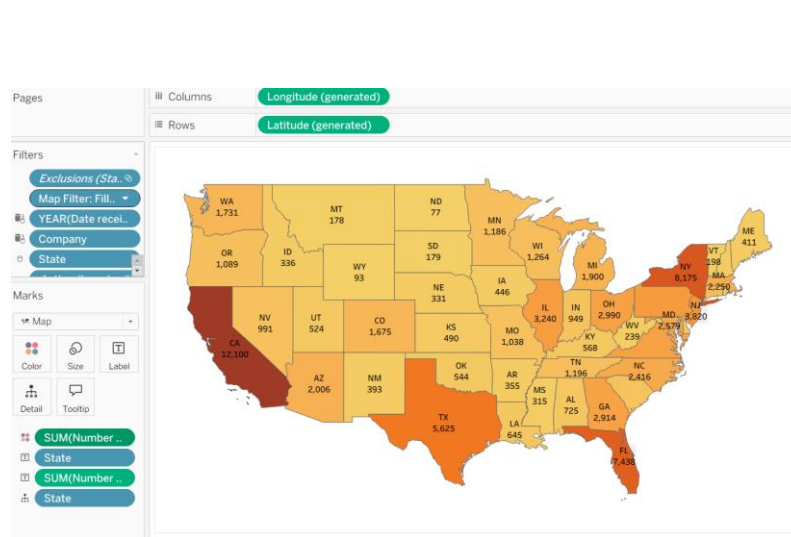
- 12.1. Used geographic fields (State, Zip) as Details.
- 12.2. Use Density as mark type.
- 12.3. Color based on intensity of complaints.



## 13. Sheet 12: Filled Map

**Objective:** Choropleth of complaints by state.

- 13.1. Used 'State' as Details on map.
- 13.2. Dragged 'State' and 'Number of Records' in Label
- 13.3. Colored by Sum of 'Number of Records'.
- 13.4. Show as filled map.



13.5. Created a parameter:

13.6. Created a calculated field:

13.7. Added Map Filter in Density Map Sheet

## 13.8. Added Map Filter in Filled Map Sheet

Filter [Map Filter]

General Wildcard Condition Top

☐ Select from list ☐ Custom value list ☐ Use all

Enter Text to Search or Add

Filled Map

Clear List ☒ Include all values when empty ☐ Exclude

Summary

Field: [Map Filter]  
Selection: Selected 1 value  
Wildcard: All  
Condition: None  
Limit: None

Reset OK Cancel Apply

## 14. Sheet 13: Top Issues

**Objective:** Show most frequent complaint types.

14.1. Used Issue on Rows and filtered

Filter [Issue]

General Wildcard Condition Top

☐ None

☒ By field

Top 10 by

Number of Records Sum

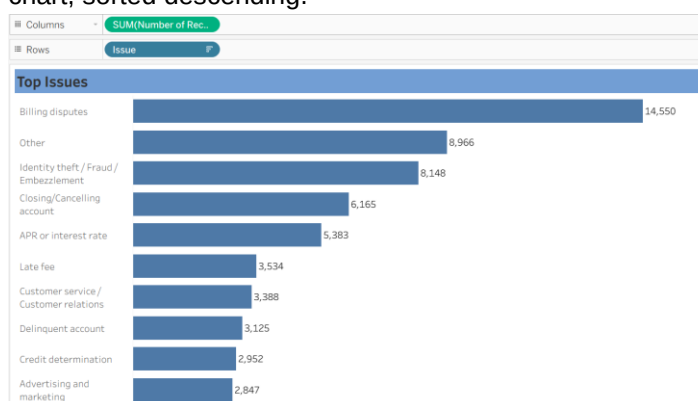
☐ By formula

Top 10 by

Reset OK Cancel Apply

14.2. Number of Records on Columns.

14.3. Horizontal bar chart, sorted descending.



## 15. Sheet 14: Company Response

**Objective:** Break down how companies responded

15.1. 'Company response to consumer' on Rows

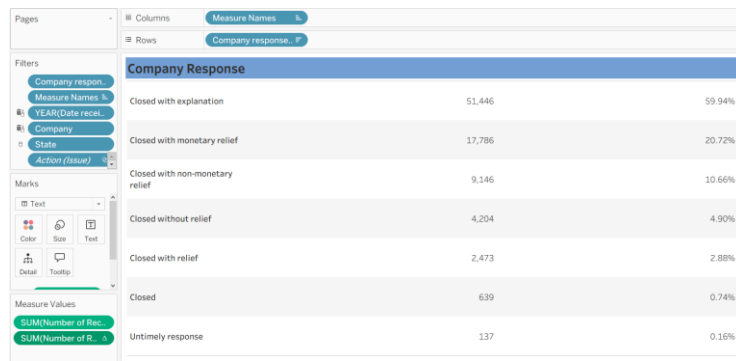
15.2. Filtered it to remove 'In Progress'

15.3. 'Number of Records' on Columns

15.4. Added 'Number of Records' to Text → Quick Table Calculation → Percent of Total

15.5. Added 'Number of Records' to Text

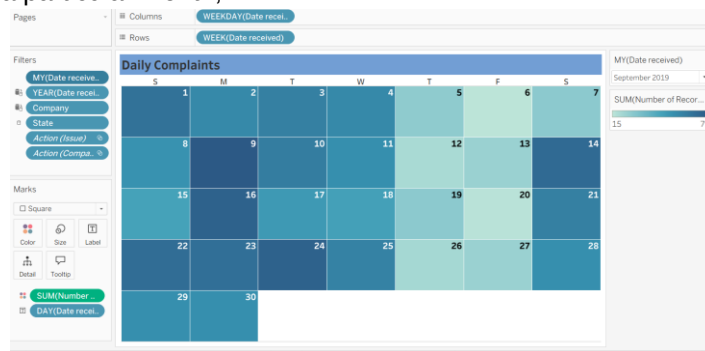




## 16. Sheet 15: Daily Complaints

**Objective:** Show daily complaints as a calendar heatmap.

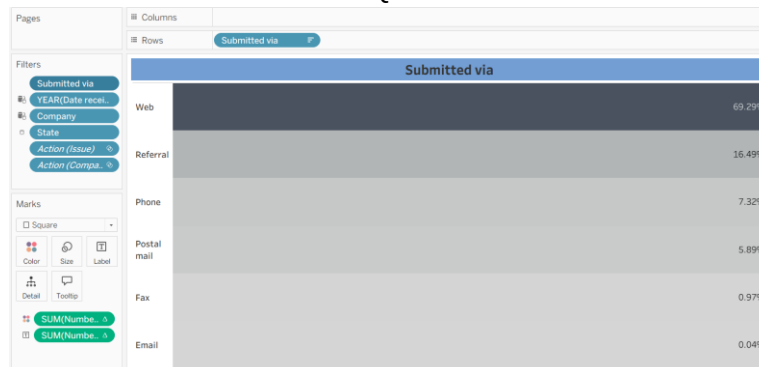
- 16.1. Used WEEKDAY(Date received) and WEEK(Date received) for axes
- 16.2. Used SUM(Number of Records) as color
- 16.3. DAY(Date received) as Title
- 16.4. Used Square as Marks
- 16.5. Filtered for a particular Month, Year of 'Date Received'.



## 17. Sheet 16: Submitted Via

**Objective:** Channel breakdown (e.g., Web, Phone).

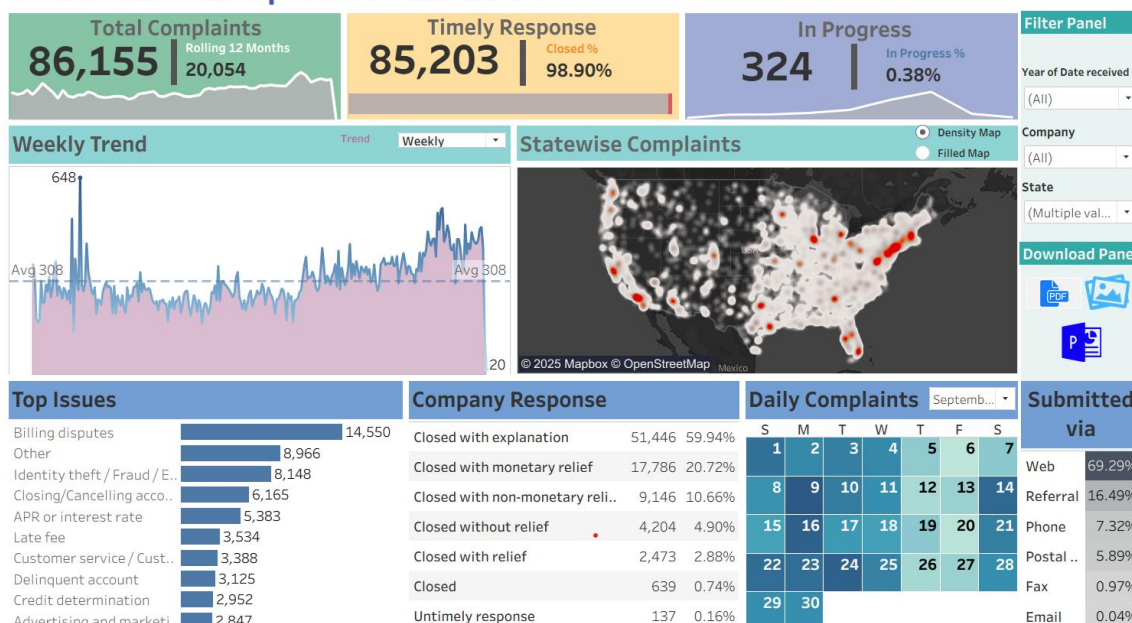
- 17.1. Used 'Submitted via' on Rows and filtered to remove Null.
- 17.2. Number of Records on Color and Label → Quick Table Calculation → Percent of Total.



## 18. Dashboard

- 18.1. Set customize size 1500X900
- 18.2. Add horizontal and vertical containers for layout and background color blocks.
- 18.3. Place KPIs as floating elements with appropriate titles and formatting.
- 18.4. Insert sparklines, maps, and other charts into fixed dashboard positions.
- 18.5. Added Filters in the Filter Panel
- 18.6. Added a section as Download Panel with the download options for pdf, image and PowerPoint.
- 18.7. Apply final formatting such as removing gridlines, adding borders, and color coding KPIs and progress bars for better visual appeal.

## Credit Card Complaints Dashboard



### Key Findings:

1. Total Complaints: 86,155 complaints, indicating significant consumer concerns.
2. High Responsiveness: 98.9% timely response rate shows excellent customer service follow-up.
3. Rolling 12 Months: 20,054 complaints, suggesting consistent ongoing issues.
4. Low Open Complaints: Only 324 complaints in progress (0.38%), implying swift closures.
5. Complaint Trends: Weekly trend shows periodic spikes, likely around billing cycles or external events.
6. Hotspot States: High complaint volumes from states like California, Texas, and New York.
7. Frequent Issues: Billing disputes, fraud, and account closures dominate.
8. Response Types: Majority resolved with explanations; very few received no relief.
9. Channels: Web submissions dominate at 69.29%.

### Conclusion:

This dashboard may enable regulators, company stakeholders, and service quality teams to understand complaint dynamics, identify service bottlenecks, and ensure swift and effective resolutions. It can empower decision-makers to:

- Identify and address key issues.
- Monitor operational performance (timely closure).
- Track trends over time and geography.
- Understand consumer behavior (submission methods, common issues).

With consistently high timely response and low in-progress complaints, companies are performing well in complaint resolution, but ongoing volume highlights the need for proactive customer engagement.